Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): Hayden Foster

Student ID: 8823130

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	2
2	10	10
3	10	10
4	10	7
5	10	9
Total:	50	38

Do not write in the table to the right.

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

   (a) The expression for the volume integral $\iiint_T f(x,y,z)\,dxdydz$ when the coordinates are changed from (x,y,z) to some other coordinates (u,v,w).
   Solution.

   $$\iiint_T f(x(u,v,w),\, y(u,v,w),\, z(u,v,w))\, dudvdw$$

   Jacobian? X

   0

   (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C.
   Solution.

   $\iint_C \vec{F}\cdot ds$ X

   Stokes: $\iint_S (\vec{\nabla}\times\vec{F})\cdot\hat{n}\,dA = \oint_C \vec{F}\cdot d\vec{r}$

   0

   (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.
   Solution.

   $\vec{N} = -2x\hat{i} - 2y\hat{j} + \hat{k}$ X

   $\vec{N} = \vec{\nabla} g$

   0

   (d) The vector field $\vec{F}$ polar coordinates written as $F_r\hat{e}_r + F_\theta\hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1\hat{i} + F_2\hat{j} = (x^2+y^2)\hat{i} - 2y\hat{j}$. *Hint:* $F_r = F_1\cos\theta + F_2\sin\theta,\ F_\theta = -F_1\sin\theta + F_2\cos\theta$.
   Solution.

   0

   (e) The expression for the divergence $\nabla\cdot\vec{F}$ in spherical coordinates (ρ,θ,ϕ) if we are given that the divergence of the same vector field is $\nabla\cdot\vec{F} = r^2 + z^2$ in cylindrical coordinates (r,θ,ϕ).
   Solution.

   2

   $r^2 = x^2 + y^2$, $\nabla\cdot F = x^2 + y^2 + z^2$

   Spherical:
   $x = \rho\cos\theta\sin\phi$
   $y = \rho\sin\theta\sin\phi$
   $z = \rho\cos\phi$

   So $\nabla\cdot F = \rho^2\cos^2\theta\sin^2\phi + \rho^2\sin^2\theta\sin^2\phi + \rho^2\cos^2\phi$

   $\nabla\cdot F = \rho^2(\sin^2\phi\underbrace{(\cos^2\theta + \sin^2\theta)}_{1} + \cos^2\phi)$

   $\boxed{\nabla\cdot F = \rho^2}$ ✓

 Points earned: 2 **out of 10 points**

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x,y,z) = x^2 + y^2 + z^2$$

in *both* Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Spherical:

$q_1 = \rho$
$q_2 = \theta$
$q_3 = \phi$
$x = \rho\cos\theta\sin\phi$
$y = \rho\sin\theta\sin\phi$
$z = \rho\cos\phi$
$h_1 = 1$
$h_2 = \rho\sin\phi$
$h_3 = \rho$

$f(\rho,\theta,\phi) = \rho^2\cos^2\theta\sin^2\phi + \rho^2\sin^2\theta\sin^2\phi + \rho^2\cos^2\phi$

$f(\rho,\theta,\phi) = \rho^2(\underbrace{\sin^2\phi(\overbrace{\cos^2\theta + \sin^2\theta}^{1}) + \cos^2\phi}_{1})$

$f(\rho,\theta,\phi) = \rho^2$

$\nabla^2 f = \frac{1}{\rho^2\sin\phi}\left[\frac{\partial}{\partial\rho}\left(\frac{\rho^2\sin\phi}{1}\left(\frac{\partial}{\partial\rho}(\rho^2)\right)\right) + \frac{\partial}{\partial\theta}\left(\frac{\rho}{\rho\sin\phi}\left(\frac{\partial}{\partial\theta}(\rho^2)\right)\right) + \frac{\partial}{\partial\phi}\left(\frac{\rho\sin\phi}{\rho}\left(\frac{\partial}{\partial\phi}(\rho^2)\right)\right)\right]$

$\frac{\partial}{\partial\theta}\rho^2 = 0$

$\frac{\partial}{\partial\rho}\rho^2 = 2\rho$

$\frac{\partial}{\partial\phi}\rho^2 = 0$

$\frac{\partial}{\partial\rho}2\rho^3\sin\phi = 6\rho^2\sin\phi$

$\frac{1}{\cancel{\rho^2\sin\phi}}\left[6\cancel{\rho^2\sin\phi}\right] = 6$

$\nabla^2 f = 6$ ✓

Cylinderical:

$q_1 = r$ $q_2 = \theta$ $q_3 = z$
$x = r\cos\theta$ $y = r\sin\theta$ $z = z$
$h_1 = 1$ $h_2 = r$ $h_3 = 1$

$f(r,\theta,z) = \underbrace{r^2\cos^2\theta + r^2\sin^2\theta}$ +

$f(r,\theta,z) = r^2 + z^2$ $x^2 + y^2 = r^2$

$\nabla^2 f = \frac{1}{r}\left[\frac{\partial}{\partial r}\left(\frac{r}{1}\left(\frac{\partial}{\partial r}(r^2+z^2)\right)\right) + \frac{\partial}{\partial\theta}\left(\frac{1}{r}\left(\frac{\partial}{\partial\theta}(r^2+z^2)\right)\right) + \frac{\partial}{\partial z}\left(\frac{r}{1}\left(\frac{\partial}{\partial z}(r^2+z^2)\right)\right)\right]$

$\frac{\partial}{\partial r}r^2 + z^2 = 2r$

$\frac{\partial}{\partial\theta}r^2 + z^2 = 0$

$\frac{\partial}{\partial z}r^2 + z^2 = 2z$

$\frac{\partial}{\partial r}(2r^2) = 4r$

$\frac{\partial}{\partial z}(2zr) = 2r$

$\nabla^2 f = \frac{1}{\cancel{r}}[4\cancel{r} + 2\cancel{r}]$

$\nabla^2 f = 6$ ✓

Points earned: 10 **out of 10 points**

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \le \rho \le 2$.

Solution.

$\phi = \text{Const} \quad d\phi = 0 \qquad ds = 1(d\rho) + \rho\sin\phi(d\theta) + \rho(d\phi)$

$dA = \rho\sin\phi\, d\rho\, d\theta$

Limits
$0 \le \rho \le 2$
$0 \le \theta \le 2\pi$
~~$0 \le \phi \le \frac{\pi}{6}$~~

$\phi = \frac{\pi}{6}, \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

$$\int_0^{2\pi}\int_0^2 \frac{1}{2}\rho\, d\rho\, d\theta$$

$\frac{1}{2} \times \frac{\rho^2}{2} = \frac{\rho^2}{4}$

$$\frac{1}{2}\int_0^2 \rho\, d\rho = \frac{\rho^2}{4}\Big]_0^2 = 1$$

$$\int_0^{2\pi} d\theta = \theta\Big]_0^{2\pi} = 2\pi$$

$= \underline{2\pi}$

Points earned: 10 **out of 10 points**

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\oiint_S \vec{F}\cdot\hat{n}\,dA,$$

and the volume integral:

$$\iiint_T (\nabla\cdot\vec{F})\,dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T.

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S, $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F}\cdot\hat{n} = (\rho\hat{e}_\rho)\cdot(\hat{e}_\rho) = \rho.$$

For $\iiint_T (\nabla\cdot F)dV$

Solution.

$x = \rho\cos\theta\sin\phi$
$y = \rho\sin\theta\sin\phi$
$z = \rho\cos\phi$

$\nabla\cdot F = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y + \frac{\partial}{\partial z}z = 3$

Limits
$0 \le \rho \le 2$
$0 \le \theta \le 2\pi$
$0 \le \phi \le \pi$
Sphere $R=2$

$dv = \rho^2\sin\phi\, d\rho\, d\theta\, d\phi$

$\int_0^{\pi}\int_0^{2\pi}\int_0^{2} 3\rho^2\sin\phi\, d\rho\, d\theta\, d\phi$

$3\sin\phi \times \frac{\rho^3}{3} = \rho^3\sin\phi$

$3\sin\phi\int_0^2 \rho^2 d\rho = \rho^3\sin\phi\Big]_0^2 = 8\sin\phi$

$\int_0^{2\pi} 8\sin\phi\, d\theta = 8\sin\phi\,\theta\Big]_0^{2\pi} = 16\pi\sin\phi$

$16\pi\int_0^{\pi}\sin\phi\, d\phi = -16\pi\cos\phi\Big]_0^{\pi} = \underbrace{[(-16\pi(-1)) - (-16\pi(1))]}_{16\pi+16\pi}$

$= \underline{32\pi}$

Points earned: 7 out of 10 points

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Continue problem 4

Solution. For $\iint_S \vec{F}\cdot\hat{n}\,dA$ $\rho = 2 = \text{Const},\ d\rho = 0$

$ds = \cancel{1d\rho} + \rho\sin\phi\, d\theta + \rho\, d\phi$

$dA = \rho^2 \sin\phi\, d\theta\, d\phi$ ✓

To find $\hat{n}$

~~$x^2 + y^2 = 4$~~

~~$\rho^2\cos^2\theta\sin^2\phi + \rho^2\sin^2\theta\sin^2\phi = 4$~~

$\rho = 2 = \text{Const},\ \boxed{n = (1,0,0)}$ X

For $\rho = \text{const}$

$\hat{n} = \hat{e}_\rho = \dfrac{1}{\sqrt{x^2+y^2+z^2}}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in $\hat{i},\hat{j},\hat{k}$ coords

in (ρ,θ,ϕ) coords

$\vec{F}\cdot\hat{n} = \rho = \sqrt{x^2+y^2+z^2}$

$F = x + y + z$

$x = \rho\cos\theta\sin\phi$

$y = \rho\sin\theta\sin\phi$

$z = \rho\cos\phi$

$F = \rho\cos\theta\sin\phi + \rho\sin\theta\sin\phi + \rho\cos\phi$

$F\cdot n = \rho\cos\theta\sin\phi(1) + \rho\sin\theta\sin\phi(0) + \rho\cos\phi(0)$ X

$F\cdot n = \rho\cos\theta\sin\phi$

$\rho = 2$

$$\int_0^{\pi}\int_0^{2\pi} \rho\cos\theta\sin\phi\,\rho^2\sin\phi\, d\theta\, d\phi = \int_0^{\pi}\int_0^{2\pi} \rho^3\sin^2\phi\cos\theta\, d\theta\, d\phi = \int_0^{\pi}\int_0^{2\pi} 8\sin^2\phi\cos\theta$$

$$8\sin^2\phi\int_0^{2\pi}\cos\theta\, d\theta = 8\sin^2\phi\sin\theta\Big]_0^{2\pi} = 0$$

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v), with transformation function

$$x = ve^{u}, \quad y = ve^{-u}.$$

Att 2:

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$\iint F \cdot n \, dA$

Parametrize

When $v = 0, u = 0$

When $v = u, u = 3$

$v = u = [1, 1, 0]$

$n = 1, 1, 0$

$F = x^2 + y^2$

$F = v^2 e^{2u} + v^2 e^{-2u}$

$\int_0^3 \int_0^3 (v^2 e^{2u} + v^2 e^{-2u}) \, dv \, du$

$e^{2u} \int_0^3 v^2 dv + e^{-2u} \int_0^3 v^2 dv$

$\left[\frac{v^3}{3} e^{2u} + \frac{v^3}{3} e^{-2u} \right]_0^3 = 9e^{2u} + 9e^{-2u}$

$9 \int_0^3 e^{2u} du + 9 \int_0^3 e^{-2u} = [$

$\det = \begin{bmatrix} \frac{\partial}{\partial v} ve^{u} & \frac{\partial}{\partial u} ve^{u} \\ \frac{\partial}{\partial v} ve^{-u} & \frac{\partial}{\partial u} ve^{-u} \end{bmatrix}$

$\underbrace{\frac{\partial}{\partial v} ve^{u}}_{e^u} \underbrace{\left(\frac{\partial}{\partial u} ve^{-u}\right)}_{-ve^{-u}} - \left(\underbrace{\frac{\partial}{\partial u} ve^{u}}_{ve^u} \underbrace{\left(\frac{\partial}{\partial v} ve^{-u}\right)}_{e^{-u}} \right)$

$-e^{u} \cdot ve^{-u} - ve^{u} \cdot e^{-u}$

$= (-2v) \underbrace{e^{u} e^{-u}}_{= 1}$

(-1)

absolute value?

$\frac{\partial(x, y)}{\partial(u, v)} = |\det D| = |-2v| = 2v$

Points earned: 9 out of 10 points

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Formula Sheet

General Coordinates:

$(x_1, x_2, x_3) \to (q_1, q_2, q_3)$:

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

Derivative Matrix: $D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$

Lamé coefficients: $h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j}\right)^2$

Jacobian: $\frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$

Line element: $ds^2 = h_1^2 dq_1^2 + h_2^2 d_2^2 + h_3^2 dq_3^2$

Volume element: $dV = h_1 h_2 h_3 \; dq_1 dq_2 dq_3$

Area element for surface with $q_j = const$ $(i, j, k$ distinct): $dA = h_i h_k \; dq_i dq_k$

Gradient for scalar function f: $\nabla f = \frac{1}{h_1}\frac{\partial f}{\partial q_1}\hat{e}_{q_1} + \frac{1}{h_2}\frac{\partial f}{\partial q_2}\hat{e}_{q_2} + \frac{1}{h_3}\frac{\partial f}{\partial q_3}\hat{e}_{q_3}$

Divergence and curl of vector field $\vec{F} = F_{q_1}\hat{e}_{q_1} + F_{q_2}\hat{e}_{q_2} + F_{q_3}\hat{e}_{q_3}$:

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}(h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2}(h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3}(h_1 h_2 F_{q_3})\right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\hat{e}_{q_1} & h_2\hat{e}_{q_2} & h_3\hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f:

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_3 h_1}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right].$$

Cylindrical Coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho\cos\theta\sin\phi, \quad y = \rho\sin\theta\sin\phi, \quad z = \rho\cos\phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho\sin\phi, \quad h_3 = h_\phi = \rho.$$